Isha Nilesh Gohel

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Education

University of Massachusetts, Amherst May 2026
MS in Computer Science (GPA: 3.77 / 4.00) *Amherst, MA*

- **Relevant Coursework:** Distributed and Operating Systems, Algorithms in Data Science, Theory and Practice of Software Engineering, Systems in Data Science, Applied Information Retrieval, Reinforcement learning, Advance Machine Learning.

Birla Institute Of Technology and Science, Pilani August 2018 – July 2023
MS in Mathematics and BE in Computer Science (GPA: 8.32 / 10.00) *Hyderabad, India*

- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Database Systems, Computer Networks, Operating Systems.

Technical Skills

Languages: Java, Python, C, C++, JavaScript, React, HTML/CSS, SQL, Bash.
Frameworks and Libraries: PyTorch, TensorFlow, OpenCV, NumPy, Pandas, Dropwizard.
Cloud and DevOps: AWS, GCP, Docker, Kubernetes, Git.
AI and ML: Generative AI, Retrieval-Augmented Generation (RAG), Prompt engineering, Fine-tuning.

Experience

Flipkart Bengaluru, India
Software Development Engineer - I *July 2023 – July 2024*

- Developed RESTful and gRPC APIs for post-order tracking features on Flipkart's health commerce platform using Java and MySQL, enabling real-time order visibility for customers and sustaining 50,000+ daily requests with high availability.
- Led the design and automation of a customer retention pipeline utilizing GCP tools (Cloud Functions, BigQuery), which directly contributed to a 22% increase in customer retention.
- Provisioned pre-production GCP environments across 6 microservices to execute NFR testing (load, performance, and reliability), ensuring API stability ahead of high-traffic sale events (Flipkart's Big Billion Days).

Software Development Intern *July 2022 – June 2023*

- Built a system to prevent Cash on Delivery (COD) abuse. Implemented per-user order limits using order APIs and risk rules, reducing fraudulent discount exploitation at scale.
- Migrated backend services from AWS to GCP by authoring Dockerfiles, Cloud Build configs, and Kubernetes manifests; deployed Java microservices on GCP using Kubernetes and integrated New Relic for performance monitoring and bottleneck identification.

Microsoft Remote
Software Engineering Mentee *June 2021 – July 2021*

- Built VCONNECT, a full-stack video calling app with real-time video, audio, screen sharing, and in-call messaging (Microsoft Engage Program).

Projects

Self-Correcting Agentic RAG | *Python, LangGraph, ChromaDB, FastAPI, Groq*

- Built a multi-agent RAG system with LangGraph orchestrating 4 LLM agents (Retriever, Answerer, Critic, Orchestrator); self-correcting retrieval via automated query rewriting and adversarial fact-checking between agents.
- 8/10 queries triggered the self-correction loop; Critic approved 6/10 answers with avg confidence 0.81 vs. no confidence scoring in naive RAG baseline.

UMass Marketplace | *FastAPI, React, MongoDB, Firebase, Gemini API*

- To give UMass students a safer alternative to Facebook groups, built a campus-exclusive full-stack marketplace featuring secure authentication, AI-assisted listing creation, real-time chat, and community events.
- Developed real-time peer-to-peer messaging to streamline transactions, ensuring a seamless user experience for the student community.

Early Breast Cancer Detection | *Python, OpenCV, Jupyter Notebook*

- Built a hybrid feature extraction pipeline from mammogram images using GLCM, Gabor, and LBP texture techniques for preprocessing and feature generation.
- Trained and evaluated multiple ML classifiers achieving 94% classification accuracy; work led to a peer-reviewed publication in Biomedical Signal Processing and Control (2023).

Distributed Image Processing System | *Python, PySpark, Hadoop HDFS, Docker*

- To classify large-scale image datasets faster, built a distributed inference pipeline across a multi-node environment, enabling parallel image processing at scale.
- Trained a custom CNN from scratch and fine-tuned ResNet50 for baseline validation, benchmarking both models across single and multi-node configurations.

Publications

- Deepa Kumari, Pavan Kumar Reddy Yannam, **Isha Nilesh Gohel**, Mutyala Venkata Sai Subhash Naidu, Yash Arora, B.S.A.S. Rajita, Subhrakanta Panda, and Jabez Christopher, **"Computational model for breast cancer diagnosis using HFSE framework"** *Biomedical Signal Processing and Control*, 86:105121, 2023